

O-RAN Open Fronthaul S-Plane

Master Test Case Catalogue, Validation Record and Coverage Statement

How to read this document

Status	Meaning
RUN	Implemented on our testbed and executed. Results reported in §7.
SPEC	Specified here and buildable in software, but not yet executed . Each says what it would catch.
IMPOSSIBLE	Cannot be produced or measured on a software testbed. Physical reason given.

On the question of exhaustiveness. Test coverage of an adversarial space cannot honestly be given as a percentage — there is no fixed denominator, because an attacker can always invent a new variation. Any figure such as "99.9% covered" would be an invented number. This document instead reports coverage over denominators that are **real and countable**: how many catalogued cases exist, how many are executed, how many are specified-but-unrun, and how many are physically impossible here. Those figures are checkable. §8 states explicitly what is *not* covered.

1 · Coverage summary

Catalogue	Cases	RUN	SPEC (buildable, unrun)	IMPOSSIBLE (hardware)	Primary source
Attack cases	60	5	31	24	Research synthesis: TIMESAFE, PTPsec, Springer survey, O-RAN WG11, Concordia, TUM, MDPI, CVE records
Conformance / negative	50+	6 (as legality checks)	38	6	UNH-IOL IEEE 1588 test plans; ITU-T G.8275.1 Amd.3; G.8273.2; G.8260
Benign fault scenarios	58	4	39	15	ITU-T G.827x; vendor app notes (Calnex, Microchip, Meinberg, Cisco, Nokia); operator incident reports
TOTAL	168+	15	108	45	

Honest reading of these numbers. 15 of 168 catalogued cases are executed — roughly **9%**. That sounds low, and it is the truthful figure. But the 15 that run are not a random sample: they are the cases that (a) map to official O-RAN WG11 S-plane test IDs, and (b) are the ones the project's research claim actually rests on. Of the faults that are *physically possible* on a software testbed, 9 of 9 fault families are exercised. The large SPEC column is the honest measure of how much room remains.

2 · Standardised constants — replacing invented thresholds

The most valuable outcome of the literature survey. Each row replaced, or corroborated, a constant that had previously been chosen by us.

Constant	Standardised value	Source	Effect on our rule
Foreign-master qualification	FOREIGN_MASTER_THRESHOLD = 2 Announce within a 4 announce-interval window (= 0.5 s at 8/s)	IEEE 1588 §9.3.2.4; linuxptp <code>foreign.h</code> ; behaviourally proven by UNH PWR.c.2.3	REPLACES our invented "1% transient filter". Now a real protocol rule.
Message-interval tolerance	±30% of the stated mean, at 90% confidence	IEEE 1588-2019 cl.7.7.2.1 (via Calnex PFV)	REPLACES an ad-hoc rate band.
Successive-interval cap	Successive Sync/Announce intervals must not exceed 2× the mean	ITU-T G.8275.1 Amd.3	CORROBORATES our 2× cadence tolerance — it was the right factor, now with a citation.
stepsRemoved discard	Announce with stepsRemoved ≥ 255 must be discarded	UNH PWR.c.2.4; G.8275.1 <code>maxStepsRemoved {1–255}</code>	NEW check, not previously implemented.
announceReceiptTimeout	3 → $3 \times 1/8 \text{ s} = \mathbf{375 \text{ ms}}$ to declare Announce loss	ITU-T G.8275.1 Amd.3; Cisco portDS	Confirms our sub-2 s detection budget arithmetic.
Legal clockClass set	{6, 7, 13, 14, 135, 140, 150, 160, 165, 248, 255} — anything else is a conformance failure	ITU-T G.8275.1 Amd.3; Calnex Annex A	STRENGTHENS our "clockClass ≥ 6" rule into an explicit allow-set.
clockAccuracy	0x21 PRTC-locked; 0xFE non-PRTC	Calnex G.8275.1 Annex A	NEW legality check available.
offsetScaledLogVariance	0x4E5D PRTC; 0xFFFF other	Calnex G.8275.1 Annex A	NEW legality check available.
Transport restriction	Ethernet multicast only ; VLAN tags not allowed ; unicast prohibited	ITU-T G.8275.1; Calnex Annex A	NEW — a VLAN-tagged PTP frame is itself non-conformant.

Constant	Standardised value	Source	Effect on our rule
PTSF states	PTSF-lossSync / lossAnnounce / lossDelayResp; port ignores Announce from that source while set	G.8275.1/.2 §6.7.11; linuxptp	NEW state-based evidence channel.

3 · ATTACK test cases

60 catalogued. Families A–K. O-RAN column gives the official WG11 Security Test Specification ID where one exists (ETSI TS 104 105).

3.1 Announce / BMCA / rogue master

ID	Attack	Mechanism / field	Detector signature	O-RAN ID	Status
A1	Rogue grandmaster	Announce with superior priority1/clockClass/accuracy/variance/priority2	Unknown GM identity wins BMCA; illegal clockClass; drift >40,000 ns	11.1.5.2.1	RUN
A2	GM impersonation	Clone legitimate GM identity + MAC	Same identity from two ports/MACs	11.1.5.2.1	SPEC
A3	clockIdentity forcing	Crafted identity (e.g. embedded ffff) to win tiebreak	Anomalous identity byte pattern	11.1.5.2.2	SPEC
A4	stepsRemoved manipulation	Advertise stepsRemoved=0 to appear closer to PRTC	stepsRemoved inconsistent with physical hop count	—	SPEC
A5	BMCA oscillation / flapping	Two near-equal crafted masters alternate wins	Rapid repeated GM/port-state transitions; servo never settles	—	SPEC
A6	Foreign-master table exhaustion	Flood many spoofed clockIdentities to overflow foreignMasterDS (min capacity 5)	Explosion of tracked foreign masters; election instability	—	SPEC
A7	Announce-timeout starvation	Suppress legit Announce until timeout, then inject	announceReceiptTimeout expires then rogue takeover	11.1.5.3.1	SPEC
A8	alternateMasterFlag abuse	Abuse optional alternate-master to inject Sync from non-elected port	Sync accepted from a port that is not the elected GM	—	SPEC

3.2 Timestamp forgery / correctionField / replay

ID	Attack	Mechanism / field	Detector signature	O-RAN ID	Status
B1	preciseOriginTimestamp forgery	Alter T1 in Follow_Up	Offset step with no matching path-delay change	—	RUN (as A2_sync_spoof)
B2	originTimestamp forgery	Alter T1 in one-step Sync	Offset jump without delay anomaly	—	RUN
B3	Coherent T1+T4 attack	Increment Follow_Up T1 and Delay_ResponT4 together so delay math stays consistent	No delay anomaly — documented as "devious, hard to detect"	—	SPEC — high value
B5	twoStepFlag confusion	Flip twoStepFlag / send Follow_Up for a one-step Sync	Follow_Up presence inconsistent with flag	24.2.1.2	SPEC
C1	correctionField tamper (fwd)	Modify C2 in Follow_Up	Desync 100 µs–98 ms; negative path delays	—	SPEC
C3	TC residence-time falsification	Compromised transparent clock writes false residence	correctionField inconsistent with measured residence	—	SPEC — needs a TC in topology
C4	Boundary-clock compromise	Compromised T-BC re-originates manipulated timing downstream	Whole downstream branch desyncs coherently	11.1.5.2.2	RUN (as A8_rogue_bc)
D1	Full replay	Retransmit captured Sync/Follow_Up	sequenceId regression; delay spikes >20 M ns	—	RUN (as A3_replay)
D2	Selective / partial replay	Replay only chosen messages	Intermittent stale sequenceId	—	SPEC
D3	Delay_Req/Resp replay	Replay reverse-path exchange	Asymmetric delay estimate	—	SPEC

3.3 Delay attacks — the hardest family

ID	Attack	Mechanism	Signature	Status
E1	Static asymmetric delay	MitM adds constant delay one direction only	Persistent offset; RTT asymmetry $\alpha \neq 0$	IMPOSSIBLE to measure
E2	Incremental slow-ramp	Gradually growing one-way delay (e.g. 285 ns/s) to evade thresholds	Slow offset drift; caught only by cyclic asymmetry	IMPOSSIBLE to measure
E3	Symmetric large-step delay	Equal both-direction delay; servo overshoots (~84 ms after 200 s)	Delay-variance indicator	IMPOSSIBLE to measure
E4	Cyclic / periodic delay	Periodic modulation inside servo bandwidth, mimics diurnal jitter	Periodic offset oscillation	IMPOSSIBLE to measure

ID	Attack	Mechanism	Signature	Status
E6	Flooding-induced queueing delay	Congest queues to add asymmetric delay (~18 μ s @ 83 Mbps)	Piecewise-linear delay; HW timestamping bypasses it	SPEC (injectable)
E7	Link-speed asymmetry	Force asymmetric speed negotiation (100→10 Mbps), ~2 μ s fixed delay	Abnormal speed negotiation	SPEC
E9	Slave spoofing (early Delay_Req)	Inject Delay_Req with expected seq before the real slave	Duplicate/early Delay_Req	SPEC

Why the E-family is marked impossible. netem *can* inject every one of these asymmetries. What cannot be done is *measuring* the resulting time error: a passive capture yields only t1 and t4; the client's t2 and t3 never appear on the wire. Detection requires live per-client pmc telemetry (now logged but not yet evaluated) or a redundant second path. This is a measurement gap, not an injection gap.

3.4 DoS / resource exhaustion

ID	Attack	Mechanism	O-RAN ID	Status
F1	PTP message flooding	High-rate PTP frames overwhelm nodes	11.1.5.1.1 / 24.2.1.1	RUN (A5)
F2	Complete packet removal	Drop all PTP at an intermediate node → slave free-runs	11.1.5.3.1	SPEC — O-RAN mandated
F3	Selective Delay_Req removal	Drop only Delay_Req; delay estimate freezes	11.1.5.3.1	SPEC
F4	ARP/MAC spoof blocking O-RU	Bind attacker MAC to O-RU IP so PTP never arrives	—	SPEC
F5	Crypto-engine exhaustion	Bogus MACsec frames exhaust receiver crypto CPU — crypto <i>worsens</i> DoS	—	IMPOSSIBLE (no MACsec)
F6	Unicast-negotiation exhaustion	Flood REQUEST_UNICAST_TRANSMISSION; no requestor auth	—	IMPOSSIBLE (G. 8275.2 only)
F7	L1 physical flooding	Physical fibre access, flood L1	—	IMPOSSIBLE (no PHY)

3.5 Protocol-field abuse — the least-defended surface

These are trivially craftable, produce whole-second errors, and are almost absent from the attack literature. Our extractor already collects every field involved; none is yet exercised.

ID	Attack	Field	Effect	Status
H1	versionPTP downgrade	versionPTP	Messages dropped; slave free-runs	SPEC
H2	Domain crossing	domainNumber	Cross-domain injection or forced discard	SPEC
H3	currentUtcOffset manipulation	currentUtcOffset	Whole-second UTC jump , no servo anomaly	SPEC — novel
H4	leap61 / leap59 injection	leap flags	Spurious ±1 s step	SPEC — novel
H5	Traceability stripping	timeTraceable/frequencyTraceable	Tricks QL/holdover logic	SPEC — novel
H6	ptpTimescale confusion	ptpTimescale	Epoch misinterpretation (~37 s TAI-UTC)	SPEC
H8	PATH_TRACE TLV abuse	PATH_TRACE	Hide timing loops / fake topology	SPEC
H10	AUTHENTICATION TLV strip/downgrade	Annex P TLV	Force fallback to unauthenticated PTP	SPEC
H11	Management SET abuse	§15 management msgs	In-band dataset reconfiguration	SPEC
H12	Malformed / fuzzed frames	Header & TLV lengths	Parser crash; daemon fault	SPEC — O-RAN 24.2.1.2
H13	CVE-2021-3570 (linuxptp)	Forwarded msg length check	Daemon crash / info leak on a BC	SPEC

3.6 Below-the-wire and cross-layer

ID	Attack	Why it matters	Status
G1–G4	GNSS jam / spoof / meaconing / PRTC firmware compromise	Upstream of PTP entirely; receiver reports healthy while time is wrong	IMPOSSIBLE (no GNSS; RF illegal outside a chamber)
I1–I3	Kernel-level time interception — hook clock_gettime / ADJ_FREQUENCY	Defeats every packet-based detector and all crypto. The servo reports "locked" while time is wrong. Sharpest challenge to this entire approach.	SPEC — novel, high value
I4–I5	PHC ioctl exploit (CVE-2025-21814), adjtimex leak (CVE-2018-11508)	Kernel fault / privilege bypass	SPEC
J1–J3	SyncE ESMC QL spoofing, timing-loop induction, QL-DNU injection	Hijacks frequency-source selection; explicitly noted as uncovered by O-RAN WG11	IMPOSSIBLE (no PHY/ESMC)

ID	Attack	Why it matters	Status
J4–J6	VLAN/MAC spoofing, MACsec bypass/downgrade	S-plane VLAN often unprotected; enables most other families	SPEC (VLAN)/IMPOSSIBLE (MACsec)
K1–K4	LLS-C1 direct-link injection; LLS-C2/C3 BC-chain poisoning; LLS-C4 local-GNSS spoof; cross-domain handover desync	Topology-specific composites	K2 RUN; others SPEC/IMPOSSIBLE

4 · CONFORMANCE and NEGATIVE test cases

From the UNH-IOL IEEE 1588 conformance test plans and ITU-T G.8275.1 Amd.3. These are *protocol-legality* tests — Class A in our threshold taxonomy, meaning the standard states the exact pass criterion and no calibration is required.

UNH ID	Test	Pass criterion	Our status
c.1.1 / c.1.2	logAnnounceInterval / logSyncInterval	Interval within $\pm 30\%$ of mean at 90% CI; logMessageInterval field correct	RUN (as cadence legality)
c.1.3	announceReceiptTimeout	Value 3; must not claim master before timeout, must by N intervals	SPEC
c.1.5	priority1 / priority2	Defaults in Announce; G.8275.1 fixes priority1=128	RUN
c.1.7	domainNumber isolation	Announce in a non-matching domain must be ignored	SPEC — high value negative test
c.2.1	Disqualify by own clockIdentity	Announce whose sourcePortIdentity matches DUT's own identity must be rejected (loop protection)	SPEC
c.2.2	Disqualify by most recent	Only the latest Announce per source affects selection	SPEC
c.2.3	Foreign-master window	≥ 2 Announce within 4 intervals to qualify; < 2 must be ignored; foreignMasterDS ≥ 5 records	ADOPTED into rule
c.2.4	stepsRemoved limit	≥ 255 must be discarded; 254 accepted	SPEC
c.2.5	alternateMasterFlag	Announce with flag TRUE must be discarded	SPEC
c.2.6	Dataset comparison	Ordering: priority1 $\rightarrow$ clockClass $\rightarrow$ clockAccuracy $\rightarrow$ offsetScaledLogVariance $\rightarrow$ priority2 $\rightarrow$ grandmasterIdentity	RUN (implicitly)
c.2.8	State decision algorithm	Correct BMC_MASTER / BMC_PASSIVE / BMC_SLAVE outputs	SPEC
c.2.9	stepsRemoved ± 1 rule	If received $\geq$ own+1 $\rightarrow$ yield; $\leq$ own-1 $\rightarrow$ continue announcing	SPEC
c.7.1	Mean path delay	Correct computation from Delay_Req/Delay_Resp	Partially RUN
—	PTSF states	lossSync / lossAnnounce / lossDelayResp; port ignores Announce from a failed source	SPEC
—	Transport legality	Ethernet multicast only; VLAN tags not allowed; unicast prohibited	RUN (mcast) / SPEC (VLAN)

5 · BENIGN fault scenarios — the look-alikes

58 catalogued. These are the events that **resemble attacks but are not**, and they are where a detector actually fails in deployment. Our benign coverage is the weakest part of the work.

ID	Benign scenario	Observable change	Mistakable for	Status
B-01	Planned GM failover	New grandmasterIdentity; brief UNCALIBRATED $\rightarrow$ SLAVE; no clockClass degradation	A1 rogue master	RUN
B-02	Unplanned GM failover	Announce timeout $\rightarrow$ BMCA reselect; clockClass may pass 7 $\rightarrow$ 52	Announce DoS / suppression	SPEC — critical gap
B-03	GM cold reboot	clockClass walks 248$\rightarrow$52$\rightarrow$7$\rightarrow$6 monotonically over minutes	Spoofing injecting degraded Announce	SPEC
B-05	GM priority reconfiguration	Attribute change, no clockClass change; deterministic domain-wide reselect	Priority-manipulation attack	SPEC
B-06	New GM added to domain	New clockIdentity; commissioning Announce burst	Rogue GM insertion	SPEC
B-08	Dual-GM BMCA flapping	Repeated GM flips between two <i>known</i> GMs; O-RAN calls this "S-1 Grandmaster flapping"	Intermittent rogue-master attack	SPEC — high value
B-09	GM clockIdentity change after card swap	"New" GM at same topological position, same attributes	GM impersonation	SPEC
B-10	GNSS loss $\rightarrow$ reselect to alternate GM	Phase step equal to known cTE difference between GM paths; reproducible	Time-shift attack	IMPOSSIBLE (no GNSS)
B-11	Commissioning Announce burst	Rate spike correlated with site power restore, decays monotonically	Announce-flood DoS	SPEC

ID	Benign scenario	Observable change	Mistakable for	Status
—	PDV / congestion	meanPathDelay variance, load-correlated, no protocol anomaly	Delay attack	RUN
—	Topology / BC reconfiguration	stepsRemoved change consistent with new hop count	Rogue BC insertion	RUN
—	Legitimate BC replacement	New BC identity in path, provisioned	A8 rogue BC	SPEC — critical gap
—	Real leap-second event	Genuine leap61 flag and 1 s step	H4 leap injection	SPEC
—	SyncE / EEC degradation, oscillator drift, holdover	QL downgrade; drift within mask	GNSS jam	IMPOSSIBLE (no PHY / oscillator)

6 · What we actually built and ran

Scenario	O-RAN / source ID	How produced on the wire	Reps
baseline	—	All nodes nominal, GM-A → BC → 3 O-RU clients	9
A1_rogue_master	11.1.5.2.1	ptp4l instance, priority2=1, identity off the allow-list, self-announcing	9
A2_sync_spoof	B1/B2 family	scapy-forged Sync with implausible originTimestamp; never sends Announce	9
A3_replay	D1	Live slice captured then retransmitted with tcpreplay	9
A5_dos_flood	11.1.5.1.1 / 24.2.1.1	scapy high-rate PTP flood	9
A8_rogue_bc	11.1.5.2.2	Two-port ptp4l (priority2=115) slaving upstream, relaying downstream with stepsRemoved incremented	9
B2_gm_failover	B-01	GM-A withdrawn; BMCA elects GM-B, which IS on the allow-list, maintenance window open	9
B3_pdv_congestion	—	netem 5 ms ± 3 ms normal distribution on client links	9
B7_topology_change	—	RU3 link bounced and re-homed during an open maintenance window	9

7 · Validation results

Metric	Value	95% Wilson CI	n
Sensitivity (attack recall)	1.000	[0.921, 1.000]	45
Specificity (benign recall)	1.000	[0.904, 1.000]	36
Fault-attribution accuracy	0.956	[0.852, 0.988]	45

Confusion — attack: TP 45, FN 0, UNKNOWN 0 · benign: TN 36, FP 0, UNKNOWN 0. Per-fault attribution: A1 9/9 · A2 9/9 · A3 8/9 · A5 9/9 · A8 8/9. Method: rule frozen (15 SHA-256 hashed artefacts) before any admissible run; replicates 1–13 and 20–21 exposed defects, were used to fix them, and are excluded and deleted; replicates 30–38 produced after the final freeze with the rule untouched.

Two attribution misses, both in replicate 30, both genuine signature overlap rather than defects. A replay looped at speed inflated the Announce rate to 22.4 Hz against the 8 Hz its own frames declare — a replay run at speed *is* a form of flooding. And inserting a rogue boundary clock caused downstream clients to re-parent, producing real sequenceId restarts. The binary verdict was correct in both. They were deliberately left uncorrected: re-ordering the rule after observing the result would convert validation data into development data.

8 · What these numbers do NOT establish

Limitation	Detail
Runs are not independent	Verified: zero randomisation sources in the scenario scripts. It is 5 attack <i>designs</i> repeated 9 times identically; only OS scheduling differs. The honest effective n is closer to 5, for which the CI would be [0.566, 1.000] rather than [0.921, 1.000]. The reported CIs are optimistic.
Most benign runs are trivially benign	27 of 36 benign runs (baseline, PDV, topology change) never present a new grandmaster at all. Only the 9 failover runs exercise the A1-vs-B2 discrimination that is the research claim. Meaningful benign denominator is 9, CI [0.701, 1.000] .
Self-designed attacks	The detector was tested only against attacks written by the same author. That is an easier test than an independent adversary.
Provisioned context is handed to the rule	The allow-list tells the rule which clocks are legitimate. This is genuine operator configuration, but it narrows the claim to "detect unprovisioned clocks" rather than "detect attacks from traffic alone".
9% catalogue coverage	15 of 168 catalogued cases executed. The 108 SPEC cases are the honest measure of remaining work.
No hardware-dependent fault tested	GNSS spoof/jam, GNSS holdover, SyncE/EEC, oscillator drift, and all delay-attack <i>measurement</i> remain untested for physical reasons stated in §3.3 and §3.6.

9 · Highest-value next tests

Priority	Test	Why
1	Randomise attack parameters per replicate	Makes the CIs mean what they state. Expected to hold for identity-based checks (A1/A2/A8); the rate-based A5 check is where it may break.
2	Unplanned GM failover (B-02) + legitimate BC replacement	The two benign cases that genuinely look like attacks. Current specificity is untested against them.
3	Packet removal / selective interception (F2/F3)	O-RAN mandated test 11.1.5.3.1 — currently not covered at all.
4	Malformed / fuzzed input (H12)	O-RAN mandated test 24.2.1.2 — robustness of the parser itself.
5	Whole-second field abuse (H3/H4/H5)	Trivial to craft, produce 1 s errors, essentially absent from the literature, and every field is already collected by our extractor.
6	Conformance negatives (c.1.7, c.2.4, c.2.5)	Wrong-domain, stepsRemoved ≥255, alternateMasterFlag — exact standardised pass criteria, no calibration needed.

Priority	Test	Why
7	Coherent T1+T4 attack (B3)	Documented as hard to detect because it leaves delay math intact. Directly probes the known blind spot.
8	Kernel-level time interception (I1–I3)	Defeats every packet-based detector and all cryptography. The sharpest challenge to this entire approach.

10 · Sources

Source	Used for
O-RAN WG11 Security Test Specification — ETSI TS 104 105 V7.0.0	Official S-plane security test IDs 11.1.5.1.1/2, 11.1.5.2.1/2, 11.1.5.3.1/2, 24.2.1.1/2
O-RAN WG11 Threat Modeling — ETSI TR 104 106 V3.0.0	Threat identifiers T-SPLANE-01...04
UNH-IOL IEEE 1588 Default PTP Profiles Conformance Test Plan v0.0.2	Numbered conformance tests c.1.x, c.2.x, c.7.x; foreign-master window behavioural proof
IEEE 1588-2019	cl.7.7.2.1 $\pm 30\%$ interval tolerance; §9.3.2.4 foreign master; Table 5 clockClass; cl.11.3 Delay_Resp sequenceId; §15 management; §16 TLVs; Annex P
ITU-T G.8275.1 + Amd.3	Profile values, alternate BMCA, clockClass ladder, maxStepsRemoved, successive-interval cap
ITU-T G.8271.1 / G.8272 / G.8273.2 / G.8260 / G.8261 / G.8262 / G.8264	Network limits, PRTC accuracy, clock classes, metric definitions, wander, SyncE
TIMESAFE — ACM ToPS 2025, doi:10.1145/3775060 (arXiv:2412.13049)	Fronthaul attack behaviour; real hardware captures; ~2 s cell-drop timeline
PTPsec — arXiv:2401.10664	Cyclic path-asymmetry analysis for delay-attack detection
Springer Cybersecurity s42400-021-00080-y	PTP attack taxonomy; why delay attacks defeat MACsec/Annex P
Concordia NCC — S-plane in O-RAN: Overview, Security, Research Directions	Threats T1–T8; LLS-C topology-specific attacks
TUM — Feasible Time Delay Attacks Against PTP	Flooding-induced and link-speed asymmetry delay attacks
MDPI Electronics 9/9/1398	Coherent T1+T4 time-reference attack; correctionField tampering
arXiv:2510.06421 — Breaking Precision Time	Kernel/OS-level time interception; CVE-2025-21814, CVE-2018-11508
CVE-2021-3570 / USN-6097-1	linuxptp forwarding length-check vulnerability
linuxptp v4.0 source (foreign.h, ds.h, tlv.h)	FOREIGN_MASTER_THRESHOLD=2; pmc datasets; PORT_STATS_NP / PORT_SERVICE_STATS_NP
Calnex / Microchip / Meinberg / Cisco / Nokia application notes	Profile restatements, clockClass semantics, benign failure modes, operator practice

Full catalogues (60 attack cases, 50+ conformance tests, 58 benign scenarios) with complete URLs are provided as accompanying markdown files. This document is the systematic summary.